



ENGINEERING MATHEMATICS-II UE22MA141B

Properties of Fourier Transform -1

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Linearity property:

If a and b are any 2 constants and $f(t)$ and $g(t)$ are any 2 functions whose Fourier transforms exist, then

$$\mathcal{F}[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)].$$

Proof: By definition of Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = \int_{-\infty}^{\infty} \{a f(t) + b g(t)\} e^{-i\omega t} dt$$

$$\mathcal{F}[a f(t) + b g(t)] = a \left[\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right] + b \left[\int_{-\infty}^{\infty} g(t) e^{-i\omega t} dt \right]$$

$$\therefore F[a f(t) + b g(t)] = a \mathcal{F}[f(t)] + b \mathcal{F}[g(t)] = a F(\omega) + b G(\omega)$$

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- Addition of two functions corresponding to the addition of the two frequency spectrum is called the linearity. If we multiply a function by a constant, the Fourier transform of the resultant function is multiplied by the same constant. The Fourier transform of sum of two or more functions is the sum of the Fourier transforms of the functions.

- **Case I.** If $h(x) \rightarrow H(\omega)$ then $ah(x) \rightarrow aH(\omega)$

- **Case II.** If $h(x) \rightarrow H(\omega)$ and $g(x) \rightarrow G(\omega)$ then

$$h(x)+g(x) \rightarrow H(\omega) +G(\omega)$$

EXAMPLE

Find Fourier transform of $y(t) = e^{-a|t|}$.

Solution

given input can be written in terms of unit step function

$$y(t) = e^{-at}u(t) + e^{at}u(-t)$$

$\mathcal{F}[y(t)] = \mathcal{F}[e^{-at}u(t) + e^{at}u(-t)]$ by Linearity property

$$= \mathcal{F}[e^{-at}u(t)] + \mathcal{F}[e^{at}u(-t)] = \frac{1}{a+i\omega} + \frac{1}{a-i\omega} = \frac{2a}{a^2+\omega^2}$$

Shifting property:

1. If a is a real constant then

$$\mathcal{F}[f(t - a)] = e^{-i\omega a} F(\omega)$$

Proof:

$$\mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

$$\mathcal{F}[f(t - a)] = \int_{-\infty}^{\infty} f(t - a) e^{-i\omega t} dt$$

Let $s = t - a \Rightarrow t = s + a$

$$ds = dt$$

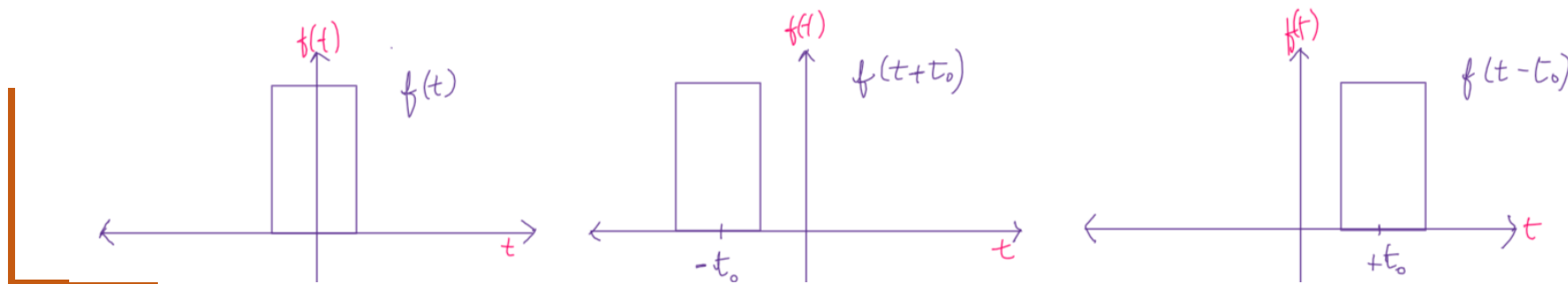
If $t \rightarrow -\infty, s \rightarrow -\infty$ and if $t \rightarrow \infty, s \rightarrow \infty$

$$\therefore \mathcal{F}[f(t - a)] = \int_{-\infty}^{\infty} f(s) e^{-i\omega(a+s)} ds$$

$$\begin{aligned} &= e^{-i\omega a} \int_{-\infty}^{\infty} f(s) e^{-i\omega s} ds \\ &= e^{-i\omega a} F(\omega) \end{aligned}$$

When the signal is shifted in time domain is said to be delayed or advanced based on whether the signal is shifted to right or left.

For example



Example

find the inverse Fourier transform of $\frac{e^{3i\omega}}{2-i\omega}$

wkt $\mathcal{F}[e^{-at}u(t)] = \frac{1}{a+i\omega}$
and $\mathcal{F}[e^{at}u(-t)] = \frac{1}{a-i\omega}$

$$\mathcal{F}^{-1}\left[\frac{1}{2-i\omega}\right] = e^{2t}u(-t) = f(t)$$

$$\mathcal{F}^{-1}\left[e^{-i\omega a}F(\omega)\right] = f(t - a)$$

$$\mathcal{F}^{-1}\left[\frac{e^{3i\omega}}{2 - i\omega}\right] = f(t + 3) = e^{2(t+3)}u(-(t + 3))$$

Frequency shifting property:

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[e^{iat}f(t)] = F(\omega - a)$

Proof:

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat} \cdot e^{-i\omega t} dt$$

$$\mathcal{F}[f(t)e^{iat}] = \int_{-\infty}^{\infty} f(t) \cdot e^{iat-i\omega t} dt = \int_{-\infty}^{\infty} f(t) \cdot e^{-i(\omega-a)t} dt$$

$$\therefore \mathcal{F}[f(t)e^{iat}] = F(\omega - a)$$

Frequency is shifted according to the co-ordinates. There is a duality between the time and frequency domains and frequency shift affects the time shift.

If $f(t) \rightarrow F(\omega)$ then $f(t) \exp[i\omega' t] \rightarrow F(\omega - \omega')$ Shifting the frequency by ω' in frequency domain is equivalent to multiplying the time domain signal by $e^{i\omega' t}$



Modulation Theorem

If $\mathcal{F}[f(t)] = F(\omega)$ and ω_0 is any real number, then

$$\mathcal{F}[f(t) \cos(\omega_0 t)] = \frac{1}{2} [F(\omega + \omega_0) + F(\omega - \omega_0)]$$

and

$$\mathcal{F}[f(t) \sin(\omega_0 t)] = \frac{i}{2} [F(\omega + \omega_0) - F(\omega - \omega_0)]$$

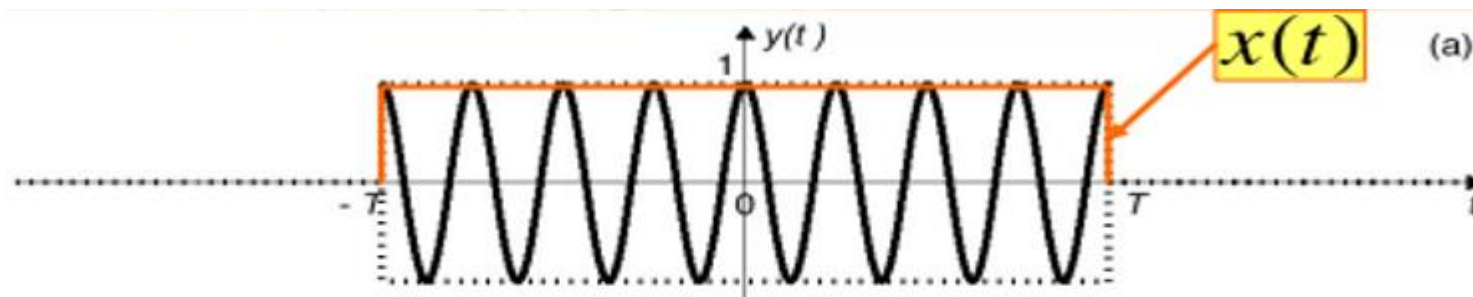
Proof w .k. t $f(t) \cos(\omega_0 t) = \frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}]$

$$\mathcal{F}\left[\frac{f(t)}{2} [e^{i\omega_0 t} + e^{-i\omega_0 t}]\right] = \frac{\mathcal{F}[f(t)[e^{i\omega_0 t}]]}{2} + \frac{\mathcal{F}[f(t)[e^{-i\omega_0 t}]]}{2} = \frac{F(\omega - \omega_0)}{2} + \frac{F(\omega + \omega_0)}{2}$$

If $\mathcal{F}[f(t)] = F(\omega)$ then $\mathcal{F}[e^{iat} f(t)] = F(\omega - a)$ (frequency shift property)

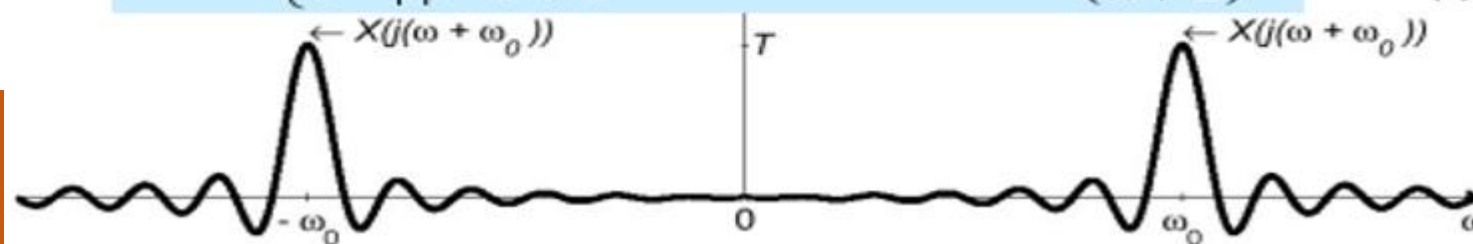
$$y(t) = x(t) \cos(\omega_0 t) \Leftrightarrow$$

$$Y(\omega) = \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$



$$x(t) = \begin{cases} 1 & |t| < T/2 \\ 0 & |t| > T/2 \end{cases} \Leftrightarrow X(\omega) = \frac{\sin(\omega T / 2)}{(\omega / 2)}$$

(b)





Thank you

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